Host (PC)

offline, into the weights:
the relabeling (L1, L3)
and the skew order (L2, L3)

per layer: im2col; window of
K = 9N taps, (s, p) → slot

skew (L2, L3): tap (s, p)
to slot s·N + (p + s) mod N

requantize the sums;
next layer

weights

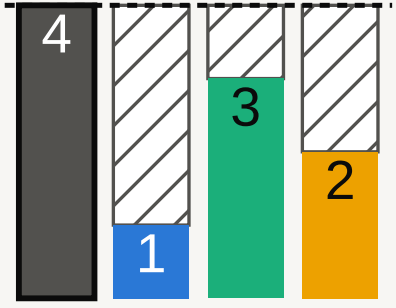
windows

sums

FPGA core (one bitstream)

window: slot j → bank j mod 4

a window of 16 taps; filled: nonzero



lockstep: all
banks wait for
the fullest, m
(hatched: idle)

each bank skips its zero taps
window: max(load, 2m, COUT), (1)